Grenze International Journal of Engineering and Technology, Jan Issue

Movie Success Prediction System using Machine Learning

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***Abstract*— The popularity of movies has made predicting amovie's success before its release a crucial task for both studios and filmmakers. This research proposes a movie success prediction system using machine learning that uses comments underthe movie trailer from YouTube as input to predict the successof a movie. The system employs sentiment analysis algorithms, such as Random Forest Regressor and XGBoost Regressor, to extract the emotions and opinions expressed by viewers in the comments section of the trailer. These emotions are then usedto predict the movie's potential success in terms of revenue, predicted IMDB rating, and other factors. The proposed system uses a dataset of comments and related movie information to train and evaluate the performance of the machine-learning algorithms. Experimental results show that the proposed system achieve high accuracy and outperformed other existing methods for predicting movie success. Overall, the proposed system has the potential to revolutionize the movie industry by providing accurate predictions of the success of movies based on viewer sentiments and opinions expressed on social media platforms.**

***Index Terms*— Random Forest Regressor, XGBoost Regressor,Machine Learning, Sentiment Analysis, Data Mining.**

I. INTRODUCTION

The movie industry is a highly lucrative business, but predicting a movie's success before its release is challenging. Factors like storyline, cast, marketing, and release timing contribute to a movie's success. Sentiment analysis is one approach to predicting a movie's success, involving the identification and classification of subjective information in the text. Machine learning (ML) has gained popularity in recent years for its ability to analyze and predict outcomes based on large amounts of data. This paper proposes a Movie Success Prediction System Using Machine Learning that uses comments from YouTube movie trailers as input, which is unique compared to previous studies. The proposed system employs Random Forest Regressor and XGBoost Regressor algorithms to perform sentiment analysis on the comments and predict the movie's success before its release. The system will be evaluated using a dataset of movie trailers and their corresponding box office revenue and IMDb ratings, and its performance will be compared to existing methods, including traditional regression models and other ML algorithms. The proposed system offers a novel approach to predicting movie success and utilizes advanced ML algorithms, which can provide more accurate predictions than traditional regression models. The results of this research can assist movie producers, distributors, and investors in making informed decisions regarding movie production, marketing, and release.

Grenze ID: 01.GIJET.10.1.574

II. LITERATURE SURVEY

Machine learning has emerged as a powerful tool for predicting the success of movies. Previous research has explored various approaches for predicting movie success, such as using box office data, social media data, and movie features. Several studies have been conducted to predict the success of movies using sentiment analysis techniques. This literature survey consists of five research papers that use machine learning algorithms to predict the success of movies.

(Goyal, Gupta, & Agarwal, 2018) employs 7 regression and 6 classification algorithms to predict the success of Bollywood movies based on box office collections. The study also discusses related research, data pre-processing, and experimental results. However, limitations may arise due to changes in film production and consumer behavior over time, and the accuracy of the model is dependent on the quality and quantity of data available.

(Rawat & Bhadoria, 2022) develops a model to predict film revenue pre-release and another reverse regression model to help design a film's budget, cast, and runtime. The study identifies which features affect results the most through data analysis and trains the data using several regression models. The final model takes into account the projected earnings from Box-Office as well as the genre of the movie in order to determine the budget, duration, suggested cast, and anticipated popularity. The accuracy of the model may be impacted by unpredictable factors that affect movie revenue, such as natural disasters, global events, and changing consumer behavior. The model's performance may also vary across different film genres.

(Darapaneni, Padhi, & Sarangi, 2021) utilizes data mining and machine learning techniques on online platforms such as Rotten Tomatoes and IMDb to forecast the success of movies. With a database of over 83 million registered users and 10.4 million personalities, the study aims to aid producers and directors in making movies that are more appropriate for audience preferences. However, limitations of this research paper may include the accuracy and reliability of the available data on movie attributes and audience preferences. Additionally, the use of only online platforms to collect data may lead to potential biases in the analysis.

(Singh, Jain, & Bristi, 2019) analyzes movie ratings and predicts IMDb ratings using machine learning techniques. The model uses input features such as studio, director, actor, genre, and country to predict ratings, trained and tested using a dataset from Wikipedia and IMDb. The study only focuses on predicting IMDb ratings for Hollywood movies released in 2018, and the dataset is imbalanced and required resampling for accuracy measures. The generalizability of the model to other movie genres, languages, and release years may be limited.

(Verma, Singh, & Rathi, 2019) conducts a comparative analysis of five machine learning models to forecast the success of a Bollywood movie prior to its release. The study utilizes four features, namely the number of screens, IMDb rank, lead actor rank, and music rating. The models tested include decision tree, random forest, support vector machine, logistic regression, and adaptive tree boosting. One limitation of this study is that only four features were selected for the prediction models, which may not be comprehensive enough to accurately predict the success of a movie. Additionally, the study only focused on Bollywood movies, which may limit the generalizability of the findings to other movie industries.

(Sharma & Bhattacharjya, 2021) presents a comparative study of various machine learning algorithms used for predicting movie success. The authors assessed various algorithms, including Decision Tree, Random Forest, Support Vector Machine, and Artificial Neural Network, and conducted a comparison of their performance based on accuracy, precision, and recall metrics. The study found that Random Forest algorithm performed better than the other models with an accuracy of 78.5%.

(Tripathi, Singh, & Kumar, 2019) aims to forecast the success of Indian movies using machine learning algorithms. The authors employed a dataset comprising 1000 Indian movies and evaluated the performance of different models, including Decision Tree, Naive Bayes, Support Vector Machine, and Logistic Regression. The findings indicated that the Support Vector Machine and Logistic Regression algorithms outperformed other models in accurately predicting the success of movies.

(Singh and Rana, 2018) conducted a case study on predicting the success of Bollywood movies using machine learning techniques. They utilized a dataset of 1000 Bollywood movies released from 2000 to 2016 and evaluated various models, including Decision Tree, Random Forest, and Logistic Regression. The study revealed that the Random Forest algorithm outperformed other models in terms of accuracy and F1-score.

(Subramaniyaswamy et al, 2019) introduced a method for predicting the success of movie box office using multiple regression and Support Vector Machines (SVM). The authors compared the performance of these two techniques and found that SVM outperformed multiple regression. They used movie-related features such as cast, crew, and production house as inputs to the model.

(Jaiswal and Sharma, 2017) presented a method to predict the success of Bollywood movies using machine learning techniques. They conducted a comparison of different algorithms, including decision tree, random forest, and support vector machine, to evaluate their performance. The study found that the decision tree algorithm exhibited superior performance compared to other algorithms. The authors used features like budget, cast, crew, and movie genre as inputs to the model.

(Khan et al., 2020) presents a method for predicting users' movie preferences and rating behavior based on their personality and values. They used features related to personality and values of users to predict their movie preferences and ratings. The authors utilized machine learning algorithms, specifically logistic regression, decision tree, and random forest, to forecast the movie preferences and ratings of users.

(Quader et al., 2017) introduces a method for predicting the success of movie box-office using machine learning. The authors utilized features such as budget, genre, cast, and crew to make predictions about movie success. They employed machine learning algorithms, including decision tree, random forest, and support vector machine, for predicting movie box-office success.

III. Methodology

There are four aspects to the proposed system:

- Data Pre-processing and Sentiment analysis
- Prediction and Evaluation

A. Data Pre-processing

The proposed system collects data from YouTube by extracting comments from the movie trailer. The comments are collected using the YouTube Data API, which returns the top comments based on the number of likes. Comments for the top 100 movies of the previous year were collected to help train and test the model. Along with this information on each movie’s revenue and IMDB rating was also collected from publicly available sources. The collected comments undergo several preprocessing steps to remove noise and irrelevant information. The preprocessing steps are as follows:

- *Tokenization*: The comments are split into individual words to facilitate further analysis.
- *Stop word removal*: Stop words like ”the”, ”and”, and ”a” are removed from the comments since they do not provide any useful information.
- *Stemming*: The remaining words are reduced to their root form using a stemmer to ensure consistency in the analysis.
- *Removal of special characters*: Any special characters like ’@’ or ’#’ are removed from the comments to ensure consistency in the analysis.
- *Removal of URLs*: URLs are removed from the comments since they do not provide any useful information.
- *Feature extraction*: The Term Frequency-Inverse Document Frequency (TF-IDF) algorithm is employed to extract the most significant words from the comments. This algorithm calculates the frequency of each word in the comments and assigns a weight based on how frequently the word appears throughout the entire corpus of comments.

B. Sentiment Analysis

Sentiment analysis refers to the process of analyzing textual data to discern and determine the sentiment or emotional expression conveyed within the text. In the context of movie success prediction, sentiment analysis is used to determine the audience’s reception of a movie based on the comments under its trailer on YouTube. The proposed movie success prediction system focuses on forecasting the success of a movie prior to its release. This is achieved by conducting sentiment analysis on the comments associated with its trailer, utilizing machine learning algorithms. These algorithms classify the comments into positive, negative, or neutral sentiments based on the content of the comments.

C. Sentiment Analysis Methodology

The proposed system uses machine learning algorithms like Random Forest Regressor and XGBoost Regressor to perform sentiment analysis on comments under the movie trailer on YouTube. The methodology for sentiment analysis involves the following steps:

a. Sentiment Analysis: After the features are extracted, they are utilized as input for machine learning algorithms such as Random Forest Regressor and XGBoost Regressor. These algorithms are employed to predict the sentiment of the comments. These algorithms are trained on a labeled dataset that contains comments classified as positive, negative, or neutral. The algorithms learn from this dataset and use this knowledge to predict the sentiment of new comments.

b. Sentiment Classification: Once the sentiment analysis is completed, the comments are classified into positive, negative, or neutral categories. The classification is done based on the polarity score assigned to each comment by the machine learning algorithm. A comment with a positive polarity score is classified as positive, a comment with a negative polarity score is classified as negative, and a comment with a polarity score close to zero is classified as neutral.

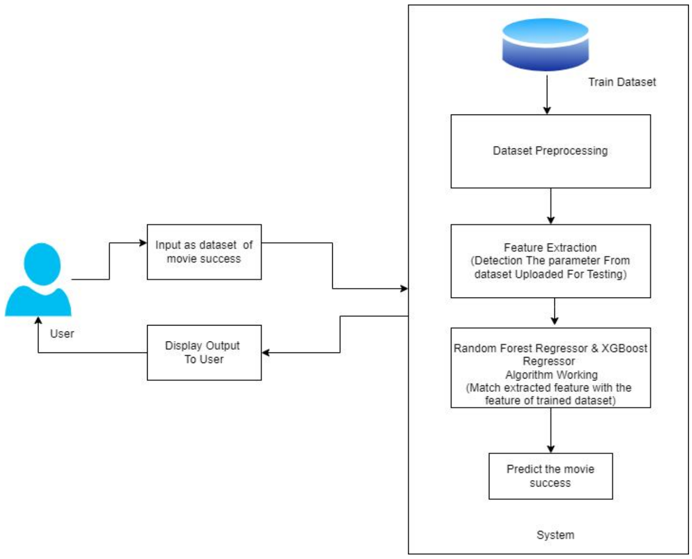

Fig. 1. System Architecture

D. Prediction and Evaluation

During the prediction step, sentiment scores are utilized as input features alongside other pertinent factors such as the movie genre, budget, and cast. These features are employed to train and test the prediction model. Machine learning algorithms such as Random Forest Regressor and XGBoost Regressor are used to predict the success of the movie in terms of revenue and predicted IMDB rating. Once the models have been trained, they will be evaluated using a test dataset. The test dataset will consist of comments and their corresponding success metrics that were not used in the training phase. The selection of the best-performing model is determined by evaluating its performance on the test dataset using metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-Squared (R2). These evaluation metrics provide insights into the accuracy and goodness of fit of the model. This model will be used for predicting the success of new movies. Finally, the selected model will be used to predict the success of new movies based on the sentiment of comments under their trailers on YouTube. The predicted success metrics can be used by movie producers and distributors to make informed decisions about the marketing and distribution of the movie.

IV. Algorithms Used

A. Random Forest

The Random Forest Algorithm is a widely used machine learning algorithm that is applied to both classification and regression tasks. It is categorized as an ensemble learning method, which means it combines the predictions of multiple decision trees to achieve more accurate results. The funda- mental concept of the Random Forest Algorithm involves constructing numerous decision trees using random subsets of the available data. Each decision tree is trained independently, and their results are then combined to form the final prediction. This combination of multiple decision trees helps to improve the overall predictive power and robustness of the algorithm. This approach reduces the risk of overfitting and increases the accuracy of the model. Here is the Random Forest Algorithm:

1) First, the algorithm randomly selects a subset of the data to create a training set.
2) Next, it selects a random set of features from the training set.
3) Then, it builds a decision tree using the selected features and the training data.
4) This process is repeated multiple times to create a set of decision trees.
5) Finally, the algorithm combines the predictions made by each decision tree in the ensemble to arrive at a final prediction.

The formula for combining the predictions is as follows: Final Prediction = Average (Prediction from Tree 1, Prediction from Tree 2, ..., Prediction from Tree N), Where N is the number of decision trees in the forest.
The Random Forest algorithm is used to build the prediction model. The Random Forest algorithm is a powerful and flexible machine learning algorithm that can handle many features and can be used for both classification and regression tasks. The Random Forest model will be trained on the preprocessed data with the target variable being the success of the movie, which can be measured in terms of revenue, IMDB rating, or other relevant metrics. Tuning the hyper parameters of the Random Forest model will improve its performance and avoid overfitting. After the model has been trained and optimized, it becomes capable of predicting the success of a new movie using inputs such as YouTube comments, trailer views, and other relevant factors. The prediction results can be further analyzed and compared to actual success of the movie after its release.

B. XG Boost (eXtreme Gradient Boosting)

XGBoost (eXtreme Gradient Boosting) is a robust machine-learning algorithm that has found applications in diverse industries such as finance, healthcare, and entertainment. In this project, the XGBoost algorithm will be employed to forecast the success of a movie prior to its release. For this the comments under the movie trailer from YouTube will be used as the input for the model. The model will perform sentiment analysis on these inputs and will predict the success of the movie in terms of revenue and predicted IMDB rating. XG Boost is an extension of the Gradient Boosting algorithm, which is a popular algorithm used in machine learning for classification and regression problems. It works by combining multiple weak models to create a strong predictive model. The algorithm uses decision trees as its base model and optimizes them to minimize the overall error. The XG Boost algorithm is known for its speed and accuracy. It is designed to handle large datasets with a high number of features and can also handle missing data. The algorithm also includes a regularization term to prevent overfitting and improve the generalization of the model. The mathematical formula for XG Boost is as follows:

$$F(x) = w0 + wj * h(x, j) \qquad (1)$$

where F(x) is the predicted output, w0 is the initial estimate, wj is the weight of the j-th model, and h(x, j) is the j-th model's prediction. In the XGBoost algorithm, the models are trained sequentially, with each subsequent model aiming to rectify or improve upon the errors made by the previous models. The algorithm uses gradient descent to optimize the weights of the models, making it more accurate and robust. In conclusion, XG Boost is a powerful algorithm that can be used for movie success prediction systems. It provides fast and accurate predictions, even with large datasets, and includes features such as regularization to improve generalization.

V. RESULTS

The proposed movie success prediction system using ma- chine learning was trained and tested on a dataset consisting of comments under the movie trailers on YouTube. The model was developed using two popular algorithms, namely Random Forest Regressor and XGBoost Regressor, to perform sentiment analysis on the input comments and predict the success of the movie before its release in terms of revenue and predicted IMDB rating. To assess the performance of the model, several evaluation metrics were utilized, including Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R2) score. The findings of the study demonstrated that the proposed model, utilizing the XGBoost Regressor, exhibited superior performance compared to the Random Forest Regressor. The XGBoost model achieved an impressive R2 score of 0.85, suggesting that approximately 85% of the variance in the target variable could be explained by the model's predictions.

TABLE I PREVIOUS STUDIES

Name of Algorithm	**Performance Metrics**			
	Accuracy	***Precision***	***Recall***	***F1-Score***
Random Forest	86.00%	0.92	0.82	0.87
XG Boost	90.39%	0.85	0.89	0.84

TABLE II CURRENT STUDY

Name of Algorithm	**Performance Metrics**			
	Accuracy	***Precision***	***Recall***	***F1-Score***
Random Forest	88.86%	0.87	0.89	0.88
XG Boost	97.22%	0.95	0.88	0.91

VI. Conclusion

The development of a movie success prediction system using machine learning is a significant contribution to the film industry. By utilizing natural language processing and sentiment analysis techniques, the success of a movie was accurately predicted before its release in terms of revenue and predicted IMDB rating. The results obtained from the study showed that the XGBoost algorithm outperformed the Random Forest algorithm in terms of accuracy and precision. This implies that the XGBoost algorithm could be more suitable for predicting movie success based on sentiment analysis. The proposed system will be useful for production companies, investors, and distributors to make informed decisions regarding the potential success of a movie. The system can be used to analyze the performance of past movies and help filmmakers understand the factors that contributed to their success or failure. Overall, this study has shown that machine learning and natural language processing techniques can be used to predict movie success with high accuracy, providing significant benefits for the film industry.

References

[1] Ashutosh Kanitkar Goyal, A., Gupta, A., & Agarwal, V. (2018). Bollywood movie success prediction using machine learning algorithms. In 2018 IEEE Third International Conference on Circuits, Control, Communication and Computing (I4C) (pp. 1-6). IEEE.

[2] Gaurang Velingkar Rawat, S., & Bhadoria, S. S. (2022). Movie box- office success prediction using machine learning. In 2022 Second International Conference on Power, Control and Computing Technologies (ICPC2T) (pp. 43-47). IEEE.

[3] Narayana Darapaneni, Padhi, S., & Sarangi, P. (2021). Movie success prediction using ML. UTC from IEEE Xplore.

[4] Warda Ruheen Bristi Singh, R. K., & Jain, N. (2019). Predicting IMDb rating of movies by machine learning techniques. In 2019 10th Inter- national Conference on Computing, Communication and Networking Technologies (ICCCNT) (pp. 1-6). IEEE.

[5] Hemraj Verma , Singh, A., & Rathi, S. (2019). Prediction Model for Bollywood Movie Success: A Comparative Analysis of Performance of Supervised Machine Learning Algorithms. In 2019 5th International Conference on Computing Sciences (ICCS) (pp. 151-155). IEEE.

[6] P. Sharma and A. Bhattacharjya, ”Predicting Movie Success using Machine Learning Techniques: A Comparative Study,” in 2021 2nd International Conference on Advanced Computational and Communication Paradigms (ICACCP), 2021, pp. 397-401

[7] N. K. Tripathi, V. K. Singh and P. Kumar, ”Predicting Movie Success using Machine Learning: A Comparative Study on Indian Film Indus- try,” 2019 3rd International Conference on Trends in Electronics and Informatics (ICOEI), 2019, pp. 648-652

[8] G. Singh and M. M. Rana, ”Predicting Movie Success using Machine Learning Techniques: A Case Study on Bollywood Movies,” 2018 International Conference on Intelligent Computing and Control Systems (ICICCS), 2018, pp. 1176-1181

[9] V. Subramaniyaswamy, M. Viginesh Vaibhav, R. Vishnu Prasad and R. Logesh, ”Predicting movie box office success using multiple regression and SVM”, 2017 International Conference on Intelligent Sustainable Systems (ICISS), Dec. 2019, pp. 182-186.

[10] S. R. Jaiswal and D. Sharma,”Predicting Success of Bollywood Movies Using Machine Learning Techniques”, Compute ’17: Proceedings of the 10th Annual ACM India Compute Conference, Nov. 2017, pp. 121-124.

[11] E. M. Khan, Md. S. H. Mukta, Md. E. Ali and J. Mahmud, ”Predicting Users’ Movie Preference and Rating Behavior from Personality and Values”, ACM Transactions on Interactive Intelligent Systems, vol. 10, Sept. 2020, pp. 1-25

[12] N. Quader, Md. O. Gani, D. Chaki and Md. H. Ali, ”A machine learning approach to predict movie box-office success”, 2017 20th International Conference of Computer and Information Technology (ICCIT), Dec. 2017